

Bond breaking and forming

Learning outcomes

By the end of this section you should be able to:

- Calculate the enthalpy change of a reaction from given average bond enthalpy data.
- Describe why bond enthalpy data are average values and may differ from those measured experimentally.

Imagine two dogs playing tug-of-war with a toy. Both dogs are 'sharing' the toy and if enough force is applied, the toy could break. Like the 'shared' toy in **Figure 1**, a chemical bond between two atoms can break if enough energy is applied. Specifically, in the case of a bond that is breaking, energy must be absorbed. Where does this energy come from? We must remember that since energy is conserved in a closed system, it must have come from the surroundings. If breaking a bond requires energy, then forming a bond must release energy to the surroundings. Can you think of a way to remember this?

Study skills

Sometimes it is helpful to think of everyday experiences as examples for remembering concepts in chemistry. For bond formation you can think of this as two magnets that are attracted to each other. Imagine holding them a short distance away from each other and then allowing the magnets to 'snap' together. The sound of them hitting each other is the energy that is released, in much the same way that energy is released when bonds are formed.



Figure 1. Two dogs playing tug-of-war.

Credit: Bill Varie, Getty Images

Use **Interactive 1** to help you remember some of these important concepts.

Bond breaking		Bond forming	

+ΔH	Energy is released	Endothermic	Energy is absorbed	-ΔH	Exothermic
-----	--------------------	-------------	--------------------	-----	------------

☒ Check

Interactive 1. Differentiating Between Bond Breaking and Formation.

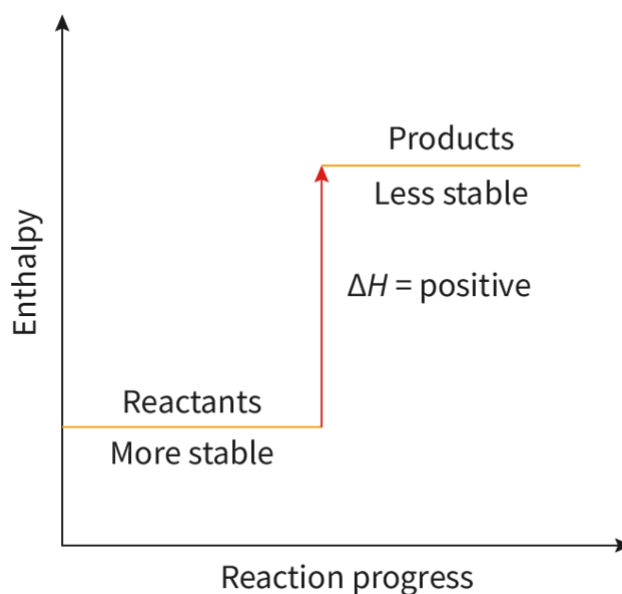
Concept

Ionic bonds are formed by electrostatic attractions between oppositely charged ions and covalent bonds are formed by the electrostatic attraction between a shared pair of electrons and the positively charged nucleus. Energy must be absorbed to break these bonds and energy is released when forming bonds.

Bond enthalpy

When we look at the equation for a chemical reaction, sometimes we overlook the steps that must take place for the reactants to be transformed into the products. Each chemical reaction involves the breaking and forming of bonds, with an overall change in energy. This change in energy reveals important information

about the stability of the reactants and products. Specifically, exothermic reactions have more stable products and endothermic reactions have less stable products compared to the reactants. This can be seen in the enthalpy diagrams in **Figure 2** where those chemical species with higher energy are less stable than those with lower energy. So, for an exothermic reaction, the products have less energy than the reactants and therefore the products are more stable (and vice versa for an endothermic reaction).



 More information

The image is an enthalpy diagram representing an endothermic reaction. The X-axis is labeled 'Reaction progress' and the Y-axis is labeled 'Enthalpy'. The diagram depicts a reaction where the reactants are located at a lower enthalpy level, labeled as 'Reactants - More stable', indicating higher stability. An arrow points upward to a higher enthalpy level labeled 'Products - Less stable', reflecting lower stability. The change in enthalpy (ΔH) is marked as positive, showing the products have higher energy than the reactants.

[Generated by AI]

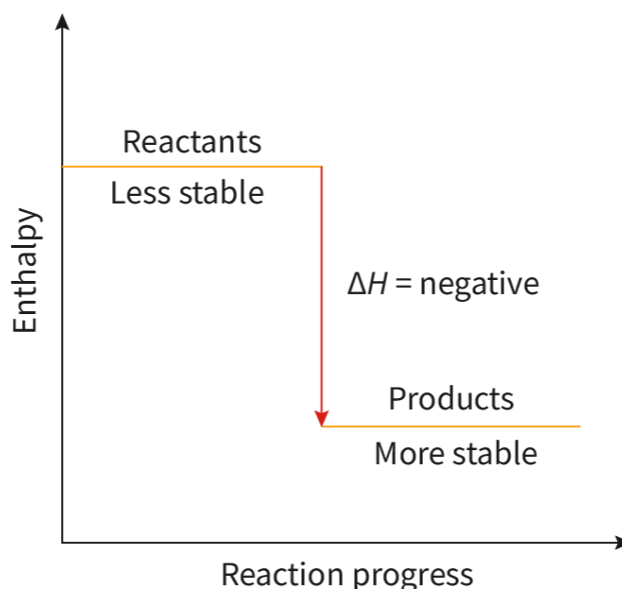


Figure 2. Endothermic and exothermic enthalpy diagrams comparing the stability of reactants and products.

 More information for figure 2

This diagram is a graph representing the enthalpy change during a chemical reaction. The Y-axis is labeled 'Enthalpy' and the X-axis is labeled 'Reaction progress.' The graph shows two horizontal lines with the upper line labeled 'Reactants' signifying 'Less stable' and the lower line labeled 'Products' indicating 'More stable.' An arrow pointing downwards between the two lines is labeled ' $\Delta H = \text{negative}$,' showing an exothermic reaction where the products are more stable than the reactants.

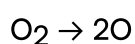
[Generated by AI]

Making connections

The relative stability of reactants and products determines whether reactions are endothermic or exothermic (see [section R1.1.3](https://app.kognity.com/:sectionlink:138726)  (<https://app.kognity.com/:sectionlink:138726>)).

If we consider this statement in relation to what we know about everyday molecules, we can reveal a useful method for remembering that bond breaking is endothermic and bond making is exothermic.

Consider the following reaction:



In this reaction, energy is required to break the double bond between the bonded oxygens, resulting in 2 oxygen atoms. But how does this relate to stability?

The form of oxygen that exists predominantly on Earth is diatomic oxygen and it is this way because this is the most stable form of oxygen. When the oxygen molecule is broken, the resulting oxygen atoms are considerably less stable. This matches **Figure 2** whereby O₂ must absorb energy in order to break the double bond.

To quantify the energy required to break a bond, we use bond enthalpy, H or bond dissociation energy, E . These terms can be used interchangeably and are defined as the energy required to break one mole of chemical bonds in the gaseous state. Bond enthalpy values are always positive because they refer to bonds being broken (bond breaking is endothermic).

Bond enthalpy values can be found in section 12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet. Examples of bond dissociation and formation are shown below. Use **Interactive 2** to investigate if they are endothermic or exothermic.



 Turn

Card 1 of 4



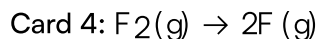
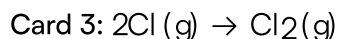
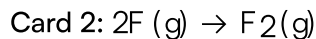
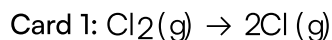
Interactive 2. Endothermic or Exothermic?

 More information for interactive 2

An interactive carousel selector features four pluck cards, each displaying a chemical reaction. Users need to identify whether the given reaction is endothermic or exothermic.

Users can switch between cards using arrows present in the bottom right and bottom left. Users can check the answer by selecting the Turn button at the bottom of each card.

The reactions given in the card are as follows:



This interactivity helps users distinguish between exothermic and endothermic reactions.

Read below for the solution:

Card 1: Endothermic

$$E(\text{Cl}-\text{Cl}) = +242 \text{ kJ mol}^{-1}$$

Card 2: Exothermic

$$E(\text{F}-\text{F}) = -159 \text{ kJ mol}^{-1}$$

Card 3: Exothermic

$$E(\text{Cl}-\text{Cl}) = -242 \text{ kJ mol}^{-1}$$

Card 4: Endothermic

$$E(\text{F}-\text{F}) = +159 \text{ kJ mol}^{-1}$$

Importantly, you need to recognise that if these reactions were reversed so that they are for the bonds forming, the enthalpy released is equal in magnitude to the energy absorbed when the bond breaks (and so we flip the sign).

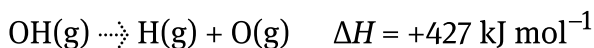
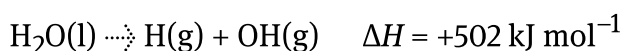
Bond enthalpies are averages

Looking at C–H bonds in organic molecules, there are many different possible combinations for the structures that can exist. Some organic molecules contain only carbon and hydrogen atoms and others contain electronegative atoms such as oxygen and nitrogen. Because the environment that the C–H bond finds itself in is different in different molecules, the bond energy will also be different. For this reason, bond enthalpies are reported as an average bond enthalpy defined as enthalpy change when one mole of bonds are broken in the gaseous state averaged for the same bond in similar compounds.

Study skills

Bond enthalpies or bond dissociation energies can be found in [section 12](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet. Note that these values are averages in units of kJ mol^{-1} .

For example, the O–H bond in water has a slightly different bond enthalpy value compared to the O–H bond in ethanol. Identical bonds in molecules with two (or more) types of bonds also have different bond enthalpy values. In water, for example, it takes more energy to break the first O–H bond than to break the second:



To calculate the average bond enthalpy for the two OH bonds in water, we take the average:

$$\begin{aligned} \text{total energy to break both bonds} &= +502 + 427 \\ &= +929 \text{ kJ mol}^{-1} \end{aligned}$$

$$\begin{aligned} \text{average bond enthalpy} &= \frac{+929}{2} \\ &= +464.5 \text{ kJ mol}^{-1} \end{aligned}$$

Compare this with the value in [section 12](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet, which is $+463 \text{ kJ mol}^{-1}$. It differs slightly because the values in the data booklet are calculated from a range of different compounds, in which the same type of bond has different bond enthalpy values.

In this definition of average bond enthalpy, the word gaseous must be emphasised. Each value that we use from [section 12](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet is for gaseous chemical species. If the molecule is in the liquid or solid phase, then even more energy will be required to break the bond since the molecule must first phase change to a gas by overcoming any forces of attraction between the particles themselves.

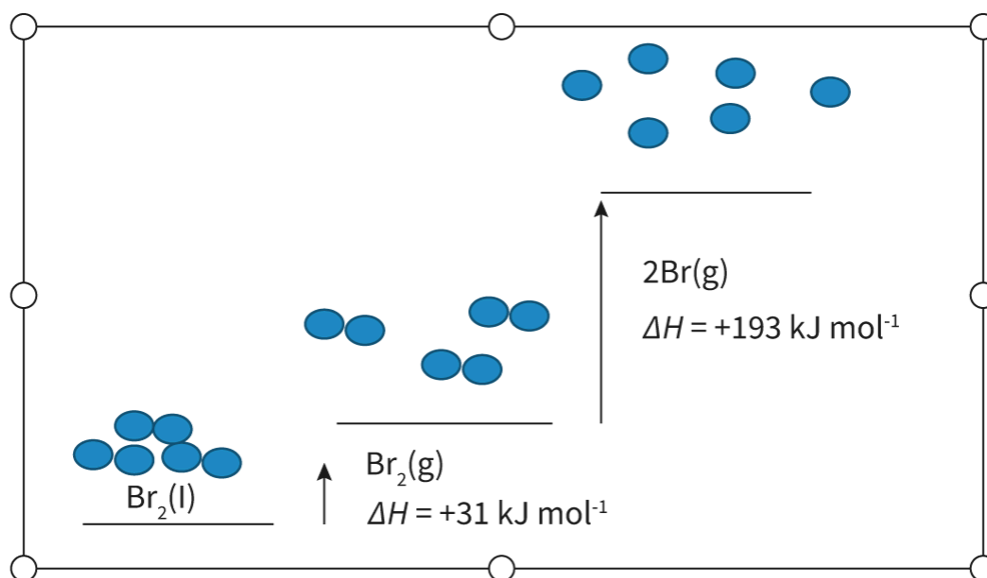


Figure 3. Energy is required to change bromine from a liquid to a gas before the bonds are broken in the gas state.

 More information for figure 3

The image is a diagram illustrating the energy changes required to transform bromine from liquid to gas and then to individual gaseous atoms. It shows three stages of bromine (Br): liquid (l), gaseous (g), and as separate bromine atoms in gas (g). At the bottom, there are colorful representations of $\text{Br}_2(\text{l})$ molecules. An upward arrow indicates a phase change to $\text{Br}_2(\text{g})$ with an enthalpy change (ΔH) of $+31 \text{ kJ mol}^{-1}$. Further above, another arrow shows the breakup of $\text{Br}_2(\text{g})$ into two bromine atoms, with a larger ΔH value of $+193 \text{ kJ mol}^{-1}$, indicating the energy required to break the bonds in the gas state. The diagram visually represents the stepwise energy increments in the transition from the liquid to the gaseous phase, ending in the dissociation of gaseous molecules into individual atoms.

[Generated by AI]

Calculate the enthalpy of a reaction using bond enthalpies

Average bond enthalpies can be used to calculate the enthalpy change of reactions. Relating back to The big picture, energy is not lost but instead changes forms. In a chemical reaction, energy is absorbed from the surroundings allowing bonds to break. Some of this energy is released again when new bonds are formed and the overall difference in energy allows us to determine whether the reaction is

exothermic or endothermic. As such, we can calculate the enthalpy of a reaction by finding the difference in bond energy between the bonds broken and the bonds formed:

$$\Delta H = \Sigma(\text{bond enthalpy (BE) of bonds broken}) - \Sigma(\text{BE of bonds formed})$$

This equation can also be considered as follows:

$$\Delta H = \Sigma(\text{reactant bond enthalpies}) - \Sigma(\text{product bond enthalpies})$$

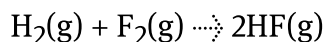
Nature of Science

Aspect: Models

Scientific models can come in a variety of forms: physical representations, abstract diagrams, mathematical equations or algorithms.

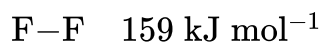
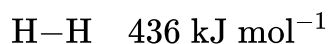
Using average bond enthalpies in the determination of enthalpy of reaction has its limitations. Since bond enthalpies are averages and are derived from a variety of molecules instead of the actual molecules in the reaction you are studying, the resulting answer can be viewed as a highly informed hypothesis as to the actual enthalpy of the reaction. If the reaction were performed in the laboratory, you would likely find that the enthalpy is very close to that which was calculated using the bond enthalpy method. It is for this reason, that modelling the enthalpy of a reaction using the bond enthalpy method of calculation is acceptable.

When determining the enthalpy of a reaction using the bond enthalpy method, it is very important that you correctly balance the reaction and determining the structural formula of each molecule in the reaction. For example, in the reaction:

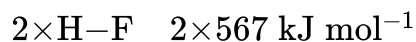


two molecules of HF are formed so the bond enthalpy of the H–F bond must be multiplied by two.

Bonds broken:



Bonds formed:



$$\begin{aligned} \Delta H &= \Sigma(\text{BE bonds broken}) - \Sigma(\text{BE bonds formed}) \\ &= (436 + 159) - (2 \times 567) \\ &= 595 - 1134 \\ &= -539 \text{ kJ mol}^{-1} \end{aligned}$$

Since the enthalpy change for this reaction is negative, it is exothermic.

When using this method, it is important to note that since average bond enthalpies are used there is some inaccuracy. While the order of magnitude of the enthalpy of reaction can be determined, experimental measurements will provide more accurate results.

In what scenarios would the bond enthalpy method be appropriate?

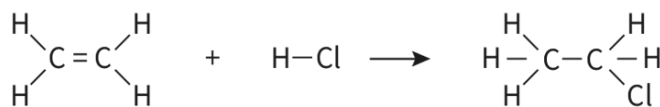
Worked example 1

For the reaction:



determine the bonds that are breaking and the bonds that are forming.

To solve questions of this type, the structural formulas of the molecules must first be determined:



Therefore, on the reactants side, the C=C bond and the H-Cl bonds are broken and on the products side, a C-H, C-C and C-Cl are formed.

Worked example 2

Calculate the enthalpy of reaction for the previous reaction using data from section 12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet.

Bonds broken:

$$1 \times \text{C}=\text{C} \quad 1 \times 614 \text{ kJ mol}^{-1}$$

$$4 \times \text{C}-\text{H} \quad 4 \times 414 \text{ kJ mol}^{-1}$$

$$1 \times \text{H}-\text{Cl} \quad 1 \times 431 \text{ kJ mol}^{-1}$$

Bonds formed:

$$5 \times \text{C}-\text{H} \quad 5 \times 414 \text{ kJ mol}^{-1}$$

$$1 \times \text{C}-\text{C} \quad 1 \times 346 \text{ kJ mol}^{-1}$$

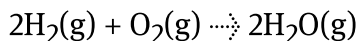
$$1 \times \text{C}-\text{Cl} \quad 1 \times 324 \text{ kJ mol}^{-1}$$

Hence:

$$\begin{aligned}\Delta H &= \Sigma(\text{BE bonds broken}) - \Sigma(\text{BE bonds formed}) \\ &= (614 + 431 + 4 \times 414) - (5 \times 414 + 346 + 324) \\ &= 2701 - 2740 \\ &= -39 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 3

Calculate the enthalpy of reaction:



using data from section 12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet.

Bonds broken:

$$2 \times \text{H-H} \quad 2 \times 436 \text{ kJ mol}^{-1}$$

$$\text{O=O} \quad 498 \text{ kJ mol}^{-1}$$

In this example it is important to note the coefficients applied to balance the equation. Since there are two hydrogen molecules, the bond enthalpy for the H–H bond must be multiplied by two.

Bonds formed:

$$4 \times \text{O-H} \quad 4 \times 463 \text{ kJ mol}^{-1}$$

As was the case for the two hydrogen molecules, the coefficient in front of the water must also be considered. Since each water molecule contains two O–H bonds and the equation is balanced with a 2 in front of the water, there are four O–H bonds formed.

Next, these values are entered into the equation:

$$\begin{aligned}\Delta H &= \Sigma(\text{BE bonds broken}) - \Sigma(\text{BE bonds formed}) \\ &= (2 \times 436 + 498) - (4 \times 463) \\ &= 1370 - 1852 \\ &= -482 \text{ kJ mol}^{-1}\end{aligned}$$

Activity

- **IB learner profile attribute:** Open-minded
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly.
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Use common objects to show the bonds that are breaking and forming in a chemical reaction to aid in solving for the enthalpy of reaction.

- Choose a simple chemical reaction and build models of each of the molecules in the reaction. Suitable materials could include toothpicks with candies/dried fruit/pieces of clay, etc. Actual molecular modelling kits could also be used.
- Create a suitable legend for the objects that you have chosen to represent each atom in the reaction.
- Take a photo of the reaction that you have modelled and use that to calculate the enthalpy of the reaction.